

Assignment for Average Students

1. Analyze the timing and power dissipation issues in FPGA-based sequential circuits. Suggest techniques to optimize performance.
2. A wearable health-monitoring device requires a processor, memory, sensors, and customized hardware acceleration. Evaluate whether an SoC, SoPC, or conventional processor-based system would be more appropriate. Justify your choice.
3. Analyze how hardware-software co-design can improve the performance of a real-time image-processing system. Identify which operations should preferably be implemented in hardware and which can be executed in software.
4. Consider a digital stopwatch as a system-level design problem. Apply the top-down design approach to divide the system into datapath, control path, and user-interface/control components. Explain the function of each part.
5. Evaluate the differences between PROM, PLA, PAL, and CPLD architectures with respect to scalability and speed.
6. An FPGA design contains several logic blocks that must communicate with each other. Analyze how programmable routing and interconnections affect routing flexibility, area, power, and signal delay.
7. Evaluate the routing architectures of Altera Stratix and Xilinx Virtex-II FPGAs for low-power applications.
8. An embedded system requires a processor and FPGA-based custom hardware accelerator. Evaluate the advantages of implementing the system as an Embedded System on Programmable Chip (SoPC). Also explain how hardware-software co-simulation can help verify the system.
9. Evaluate different programmable interconnection schemes and their impact on power dissipation in FPGA.
10. Design a hardware-software co-design strategy for an FPGA-based vending machine controller. Include the partitioning of tasks between hardware and software.